

Image Reconstruction Using fMRI Data: Improving VAE-VQGAN Implementations via Denoising and TV Filtering

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TITLE CHANGE

Scope Shift Towards Concrete, Reproducible Goals

- Early replication challenges and environment compatibility issues (outdated dependencies, lack of pre-trained weights) led to a shift in focus toward reproducibility and improvement of existing architectures.
- Rather than attempting infrastructure-heavy setups, we prioritized quality enhancement through better post-processing.

Contributions More Aligned with Image Quality Improvements

- Our actual contribution lies in improving existing reconstruction pipelines (Shen, Kamitani, VCGAN) through noise suppression techniques:
 - Denoising algorithms for raw image enhancement.
 - Total Variation (TV) filtering for preserving structure and reducing artifacting.
- These techniques led to measurable improvements in SSIM (Structural Similarity Index Measure), PSNR (Peak Signal-to-Noise Ratio), and MSE, making the updated focus more accurate and technically meaningful.

Title Alignment with Deliverables and Evaluation Metrics

- The new title reflects what was actually implemented, validated, and evaluated.
- Maintains academic clarity and ensures that reviewers and instructors can accurately assess the core contribution: improved image fidelity in brain decoding pipelines.

PREMINOR REVIEW

Comprehensive survey of deep learning approaches for fMRI-based natural image reconstruction:

- **Shen et al. (2019) – Feature-based reconstruction using VGG/DGN.**
- **Kamitani Lab Pipelines – Two-stage feature decoding and inversion.**
- **Koide-Majima et al. (2024) – Semantic-guided estimation using CLIP.**

Additional models: BeliyEncDec, FangSSGAN, QiaoGAN, and BigBiGAN.

Preprocessing Steps:

- **Trial averaging of test fMRI responses to improve SNR.**
- **Voxel selection using PCA + correlation filtering for dimensionality reduction.**
- **Matched each fMRI sample with corresponding stimulus image, resized to 224×224 (VGG/VQGAN input size).**
- **Converted fMRI activations to flattened vectors for model input.**

Evaluated each model based on:

- **Architectural complexity**
- **Dataset and training requirements**
- **Public code availability and reproducibility**

Identified key gaps:

- **Difficulty in reconstructing imagined/abstract images using only low-level features.**
- **Poor generalizability across subjects and semantic categories.**

MAJOR CHALLENGES FACED

INFRASTRUCTURE & RESOURCE CONSTRAINTS

- **NO HPC PROJECT ID: WE LACKED AN OFFICIAL PROJECT ID REQUIRED FOR TRAINING JOBS ON THE HIGH PERFORMANCE COMPUTING (HPC) CLUSTER.**
- **STANDARD QUEUE FAILURES: DESPITE MULTIPLE FIXES AND HELP FROM CSC, JOBS FAILED TO EXECUTE CORRECTLY, STALLING PROGRESS SIGNIFICANTLY.**

LOCAL SETUP CHALLENGES

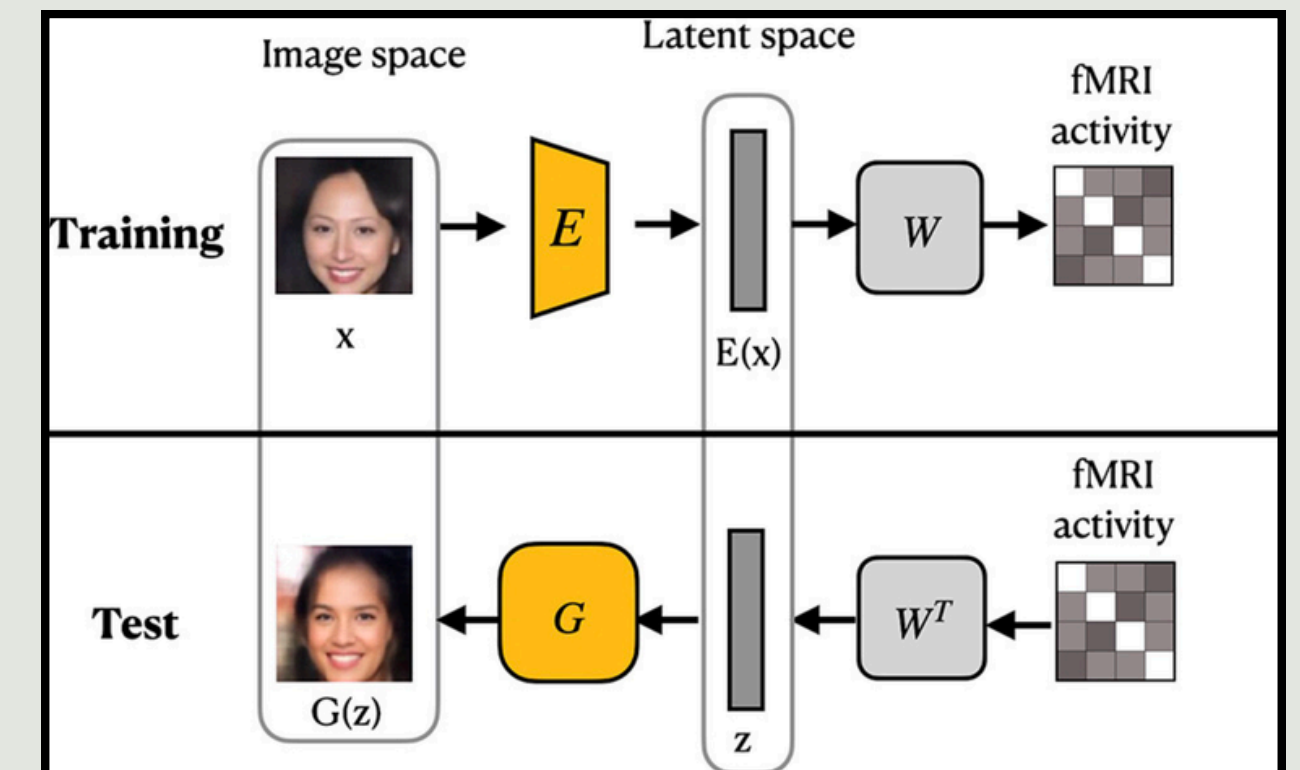
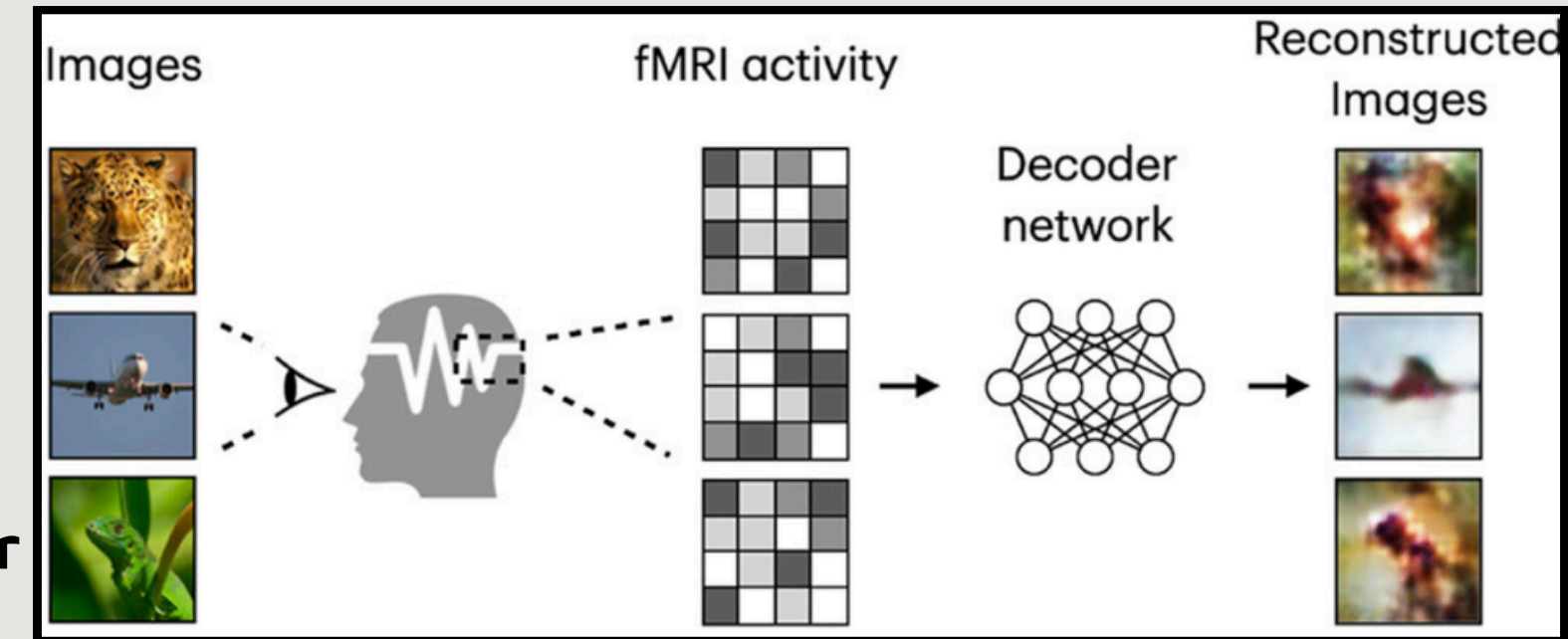
- **LARGE DATASET HANDLING: THE DATASETS WERE TOO LARGE FOR INTERNAL STORAGE, LEADING US TO PURCHASE AN EXTERNAL SSD FOR FASTER LOCAL ACCESS AND PROCESSING.**
- **LONG DOWNLOAD TIMES: COMPLETE DATA ACQUISITION TOOK OVER A DAY, ADDING TO DELAYS.**

LEGACY CODE & DEPENDENCY ISSUES

- **OUTDATED GITHUB REPOSITORIES: MANY RECONSTRUCTION MODELS (ESPECIALLY SHEN ET AL.) WERE BUILT 6–7 YEARS AGO WITH DEPRECATED DEPENDENCIES.**
- **INCOMPATIBILITY WITH MODERN SYSTEMS: MACOS AND NEWER LINUX ENVIRONMENTS NO LONGER SUPPORT MANY REQUIRED PACKAGES, MAKING REPLICATION PAINFUL.**
- **EXTENSIVE DEBUGGING REQUIRED: RESOLVING COMPATIBILITY ISSUES AND RECONFIGURING ENVIRONMENT VARIABLES TOOK A LARGE PORTION OF THE PROJECT TIMELINE.**

LITERATURE INSIGHTS

- **Gallant et al. (2008, 2009):** Demonstrated early success in reconstructing visual stimuli from individual-specific fMRI data.
- **Encoding/Decoding Models:** Naselaris et al. (2011) emphasized Bayesian decoding for reconstructing natural images.
- **Deep Learning Advances:** Recent models leverage CNNs and GANs for improved image reconstruction accuracy (Rakhimberdina et al. 2021, Shen et al. 2019, VanRullen et al. 2019).
- **Hierarchical Feature-Based Reconstruction:** Shen et al. (2019) introduced a method to reconstruct natural images from fMRI by decoding hierarchical deep features (from VGG19) and using a GAN-based generator to produce realistic images from brain activity.
- **Rakhimberdina, Z., Jodelet, Q., Liu, X., & Murata, T. (2021).** Natural image reconstruction from fMRI using deep learning: A survey. *Frontiers in Neuroscience*, 15, 795488.



VanRullenVAE-GAN framework proposed by VanRullen and Reddy (2019) Page 5

MODEL REVIEW

Shen 2019 (DNN + DGN)

Approach:

Translates fMRI voxel patterns into hierarchical deep neural network (VGG19) feature activations.
Optimizes a pixel image such that its DNN features match the decoded features from brain activity.

Architecture:

Uses a pretrained VGG19 for feature extraction across 8 layers.
Optional use of a Deep Generator Network (DGN) to improve realism in output images.

Dataset Used:

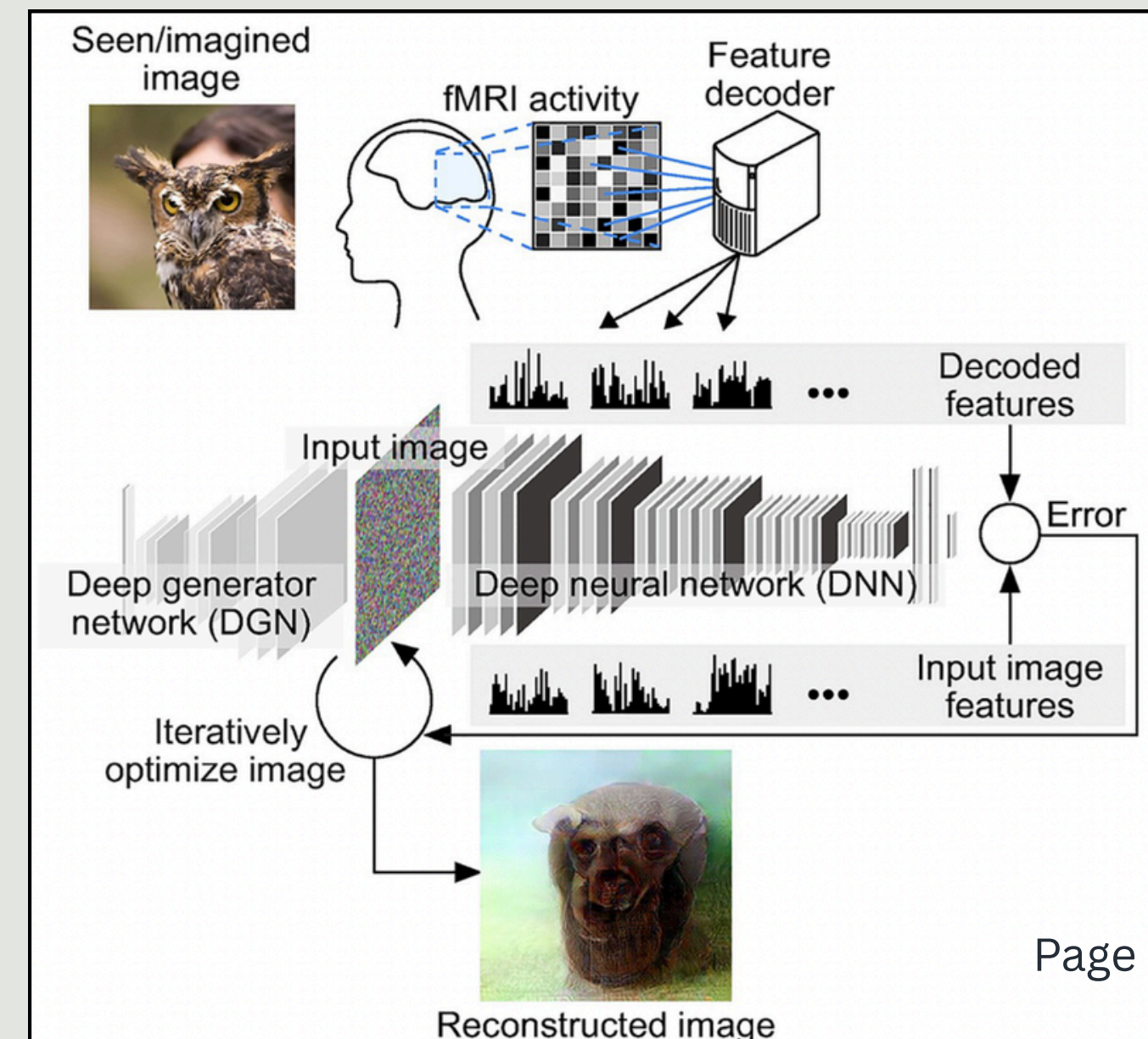
- fMRI data from 3 subjects viewing 1200 natural images (ImageNet-derived).
- 50 test images (natural), 40 artificial shapes, and imagery tasks.
- Publicly available: [Shen et al. dataset on Figshare](#).

Training:

Sparse linear regression decodes each VGG layer's features from fMRI data.
Pixel optimization via L-BFGS to minimize distance between real and decoded features.

Highlights:

Combines low- to high-level visual features across DNN layers.
Outputs visually plausible natural images and geometric patterns.



MODEL REVIEW

Kamitani Lab Two-Stage Pipeline – Feature Decoding + Image Generation

Approach:

- **Step 1: Decode DNN features (e.g., AlexNet/VGG) from fMRI using regression.**
- **Step 2: Reconstruct image via feature inversion or pretrained generator (e.g., Deep Generator Net).**

Architecture:

- **Regression from fMRI to mid-/high-level CNN features.**
- **Reconstruction via image optimization or pretrained GAN inversion.**

Dataset Used:

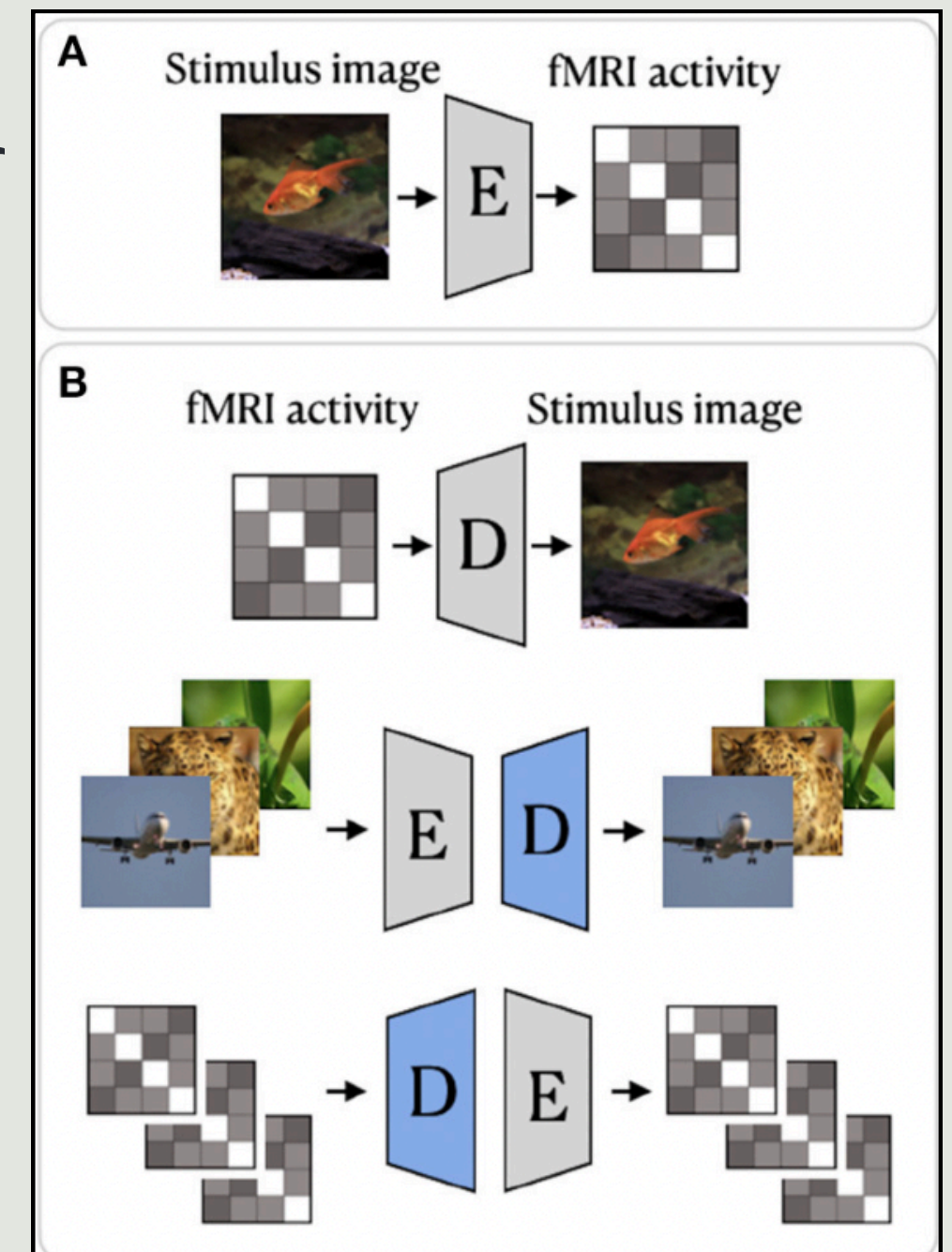
- **Data from Kamitani Lab's early visual decoding experiments.**
- **Same base structure as Shen's dataset: 1200 training images, 50 natural test images.**
- **Often cross-referenced with studies like Horikawa & Kamitani (2017).**

Training:

- **Ridge regression per feature layer; inversion uses optimization or generator latent manipulation.**

Highlights:

- **Interpretability across layers; layer-wise contribution analysis possible.**



MODEL REVIEW

Bayesian + Semantic Extension of Shen

Approach:

- Extends Shen's method using Bayesian estimation with semantic guidance from CLIP.
- Samples images from a posterior distribution using SGLD.

Architecture:

- Combines VGG19 + CLIP (ViT-B/32) feature decoding from fMRI.
- Uses VQGAN as pretrained image generator for prior distribution.
- Reconstruction guided by visual and semantic likelihoods.

Dataset Used:

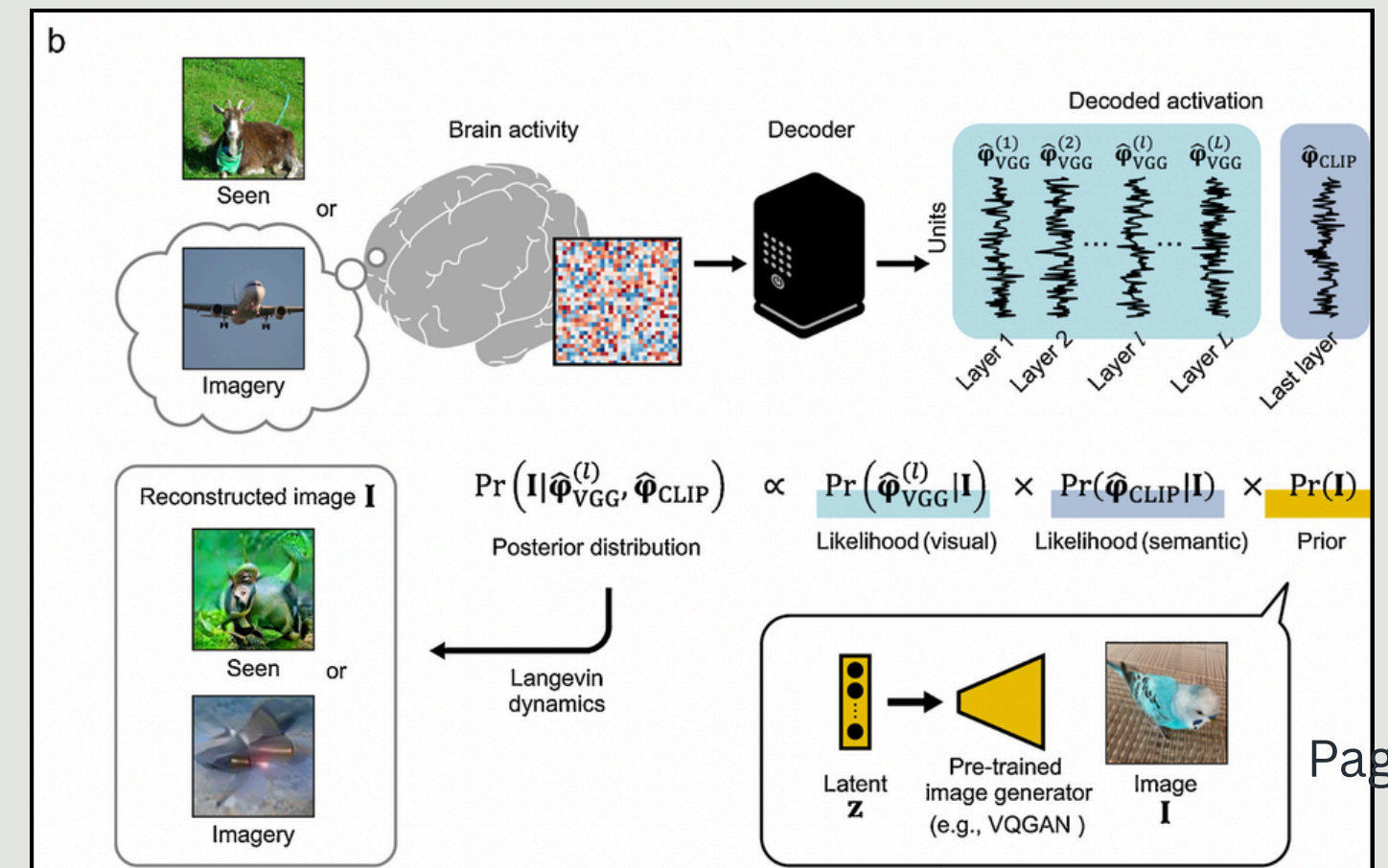
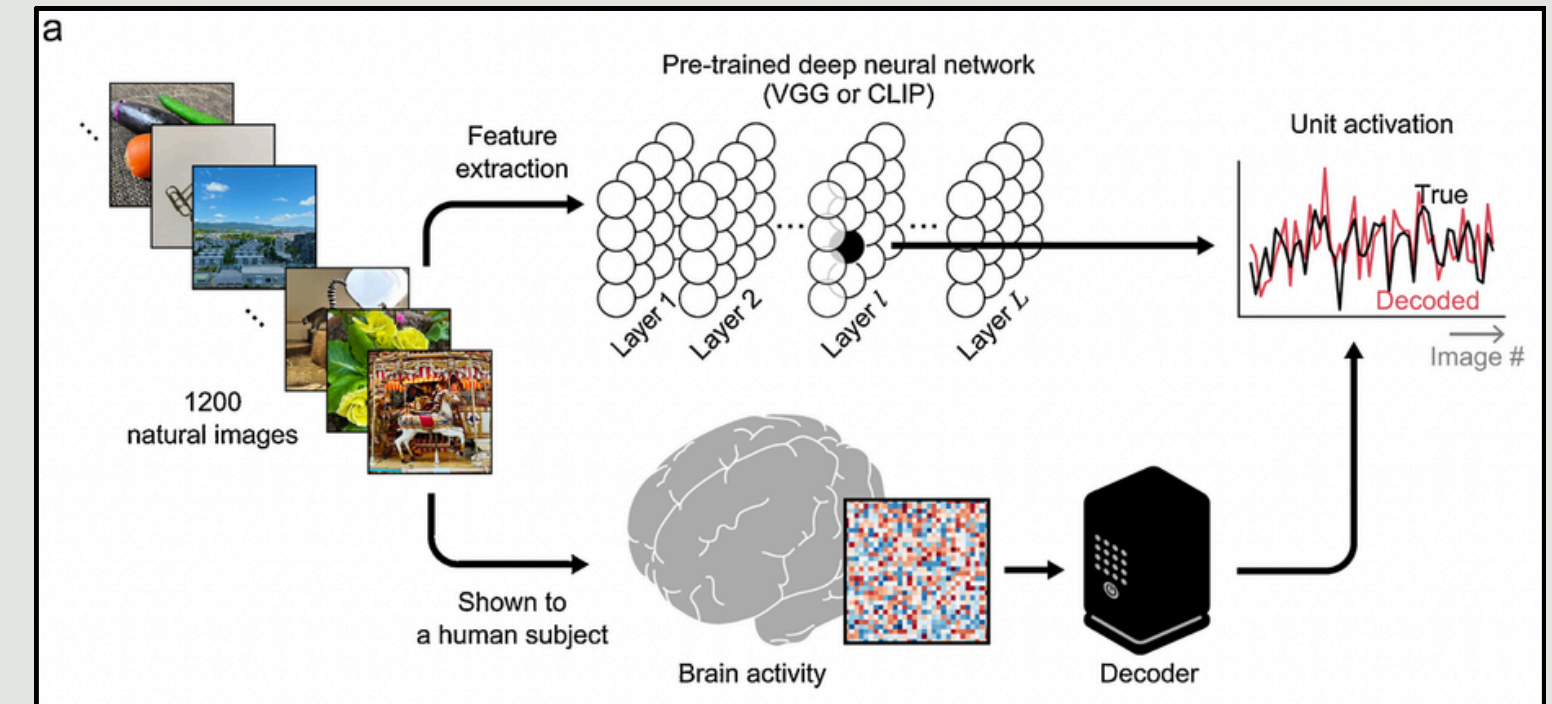
- Same fMRI dataset as Shen et al. (2019), reused under identical preprocessing.
- 1200 train images; 50 test natural, 40 artificial; imagery of 25 images.

Training:

- Linear regression with PCA-based voxel selection.
- CLIP and VGG features decoded separately and jointly used in reconstruction.

Highlights:

- Enables high-quality reconstruction from both seen and imagined images.



MODEL REVIEW

I BeliyEncDec (Beliy et al., 2019)

- **Method:** CNN-based encoder–decoder trained using supervised + self-supervised learning.
- **Reason for Not Implementing:**
 - Requires both image $\rightarrow$ fMRI and fMRI $\rightarrow$ image mappings with dual training loops.
 - Heavy reliance on self-supervised pretraining using large unlabeled datasets.
 - Original codebase not fully reproducible and performance highly sensitive to training dynamics.
 - Reported to produce blurry reconstructions compared to newer GAN/VAE-based methods

II RenD-VAE/GAN (Ren et al., 2021)

- **Method:** Dual-encoder (visual and cognitive) with knowledge distillation and GAN-based decoder.
- **Reason for Not Implementing:**
 - Involves three-stage training, including knowledge distillation from image encoders to fMRI decoders.
 - Demands large auxiliary datasets and computational resources.
 - Designed to decode abstract cognitive latent variables—not directly optimized for direct fMRI-to-image reconstruction.

III SeeligerDCGAN (Seeliger et al., 2018)

- **Method:** Uses a DCGAN with linear mapping from fMRI $\rightarrow$ GAN latent space.
- **Reason for Not Implementing:**
 - Designed for low-res grayscale images (64x64); limited relevance to your ImageNet-scale reconstructions.
 - Linear mapping severely limits expressivity; newer nonlinear GANs outperform.
 - Code not maintained; incompatible with modern libraries.

DATASET

I Kamitani Lab Dataset (Deep Image Reconstruction)

- **Source:** Kamitani Lab, Kyoto University
- **Size:** Approximately 219 GB
- **Subjects:** 3
- **Stimuli:** 1290 (ImageNet + shapes) (natural images)
- **Data Provided:**
 - **Preprocessed fMRI Data:** Voxel-wise responses to visual stimuli
 - **Decoded DNN Features:** Features from various layers of deep neural networks decoded from fMRI data
 - **.mat Files:** Contain structured data including fMRI responses and decoded features
- **Applications:** Facilitates studies on reconstructing visual images from brain activity by providing both neural responses and corresponding decoded features

II BOLD5000 Dataset

- **Source:** Carnegie Mellon University
- **Size:** Approximately 219 GB
- **Subjects:** 4 participants
- **Stimuli:** 5,254 images from SUN, COCO, and ImageNet datasets
- **Data Provided:**
 - **Raw fMRI Data:** DICOM and NIfTI formats
 - **Preprocessed fMRI Data:** BIDS-formatted data with motion correction, normalization, and other preprocessing steps
 - **Behavioral and Physiological Data:** Includes participants' responses and physiological measurements during scanning
 - **ROI Masks and GLM Analyses:** Region of interest masks and general linear model analyses for various brain regions
- **Applications:** Supports research on visual perception, neural representation of complex scenes, and development of brain-computer interfaces

EVALUATION METRICS

MSE (Mean Squared Error)

- **Type:** Pixel-wise loss
- **Definition:** Average of the squared differences between corresponding pixels of the original and reconstructed images.
- **Interpretation:** Lower MSE = better reconstruction.
- **Limitations:**
 - Penalizes even small shifts or natural variations in GAN outputs.
 - Often used during training but not ideal for final perceptual evaluation.

SSIM (Structural Similarity Index Measure) [Shen Reconstruction Implement]

- **Purpose:** Measures perceptual similarity by comparing local patterns of pixel intensities that have been normalized for luminance and contrast.
- **Why we used it:**
 - Sensitive to perceptual features such as texture, edge sharpness, and contrast.
- **Interpretation:**
 - Ranges from 0 to 1, where 1 indicates perfect similarity with the ground truth | Robust to small image transformations.

PCC (Pearson Correlation Coefficient) [Semantic Implement]

Purpose: Measures linear correlation between the ground-truth image & the reconstructed image on a pixel or feature level.

Why we used it (as in Koide-Majima et al., 2024):

Used directly as the similarity function in the Bayesian posterior log-likelihood during reconstruction.

Reflects voxel-level decoding accuracy and is directly tied to reconstruction performance.

Interpretation:

Values range from -1 (perfect negative correlation) to 1 (perfect positive correlation) | Higher PCC indicates better alignment

RESULTS FROM END2END DEEP RECONSTRUCTION (KAMITANI LAB)

- **MODEL: DIRECT FMRI → IMAGE MAPPING (CAFFE, NO INTERMEDIATE FEATURE DECODING)**
- **DATA: VISUAL CORTEX (VC), NATURAL IMAGE STIMULI, 1 SUBJECT USED ONLY**
- **LIMITATIONS:**
 - **NO SEMANTIC LAYER USAGE**
 - **CONSTRAINED FMRI DATA SCOPE (VC ONLY)**
 - **LEGACY MODEL (CAFFE + SMALL SAMPLE SIZE)**

Image ID	Original Class	SSIM	PCC	Comments
n039594393_10038	Coin-like object	0.171	0.072	Weak structural resemblance
n04572121_3262	Welding helmet	0.120	0.172	Very faint silhouette of object
n03716966_28524	Stringed musical instrument	0.133	0.441	Texture artifacts, slightly better PCC
n01677366_18182	Lizard (green background)	0.062	0.276	High noise, minimal structure retained

REPLICATION RESULTS - 1ST WORKING MODEL

Original n03954393_10038



Reconstructed
SSIM=0.171, PCC=-0.072



Original n04572121_3262



Reconstructed
SSIM=0.120, PCC=0.172



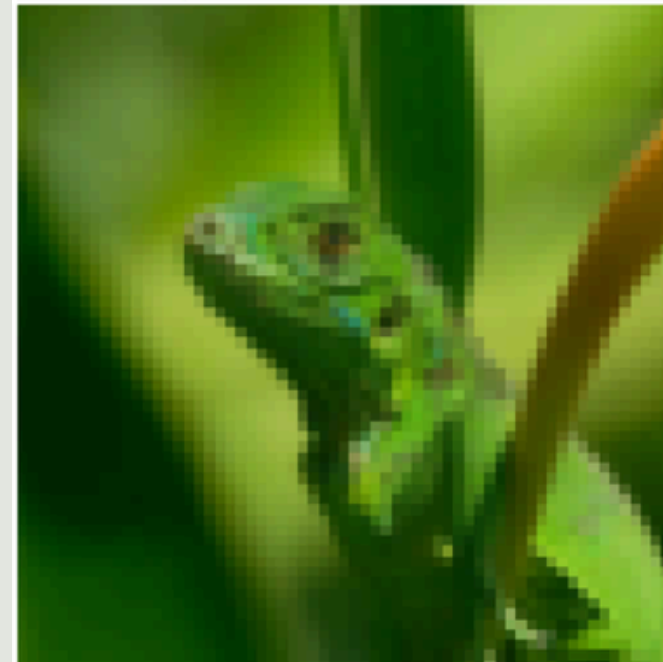
Original n03716966_28524



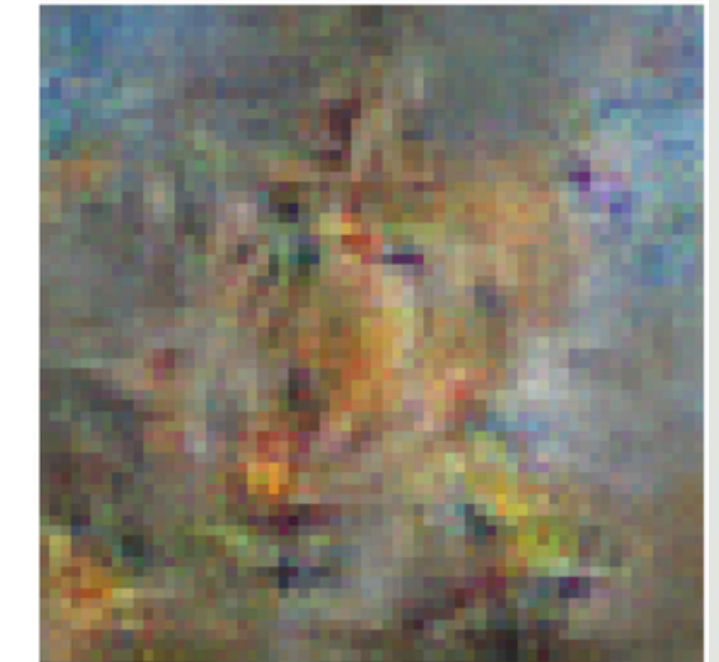
Reconstructed
SSIM=0.133, PCC=0.441



Original n01677366_18182



Reconstructed
SSIM=0.062, PCC=0.276



LOW-LEVEL DEEP IMAGE RECONSTRUCTION (SHEN ET AL. 2019 REPLICATION)

- **OBJECTIVE:**

- RECONSTRUCT VISUAL IMAGES FROM FMRI BY OPTIMIZING AN IMAGE SUCH THAT ITS VGG19 FEATURES MATCH BRAIN-DECODED DNN FEATURES.

- **KEY DETAILS:**

- INPUT: FMRI-DECODED VGG FEATURES (PRECOMPUTED)
- MODEL: GRADIENT-BASED PIXEL OPTIMIZATION USING VGG19
- GENERATOR (RELU7GENERATOR): USED FOR HIGH-LEVEL RECONSTRUCTION (FC7)
- DATA: NATURAL IMAGES FROM IMAGENET
- FRAMEWORK: PYTORCH

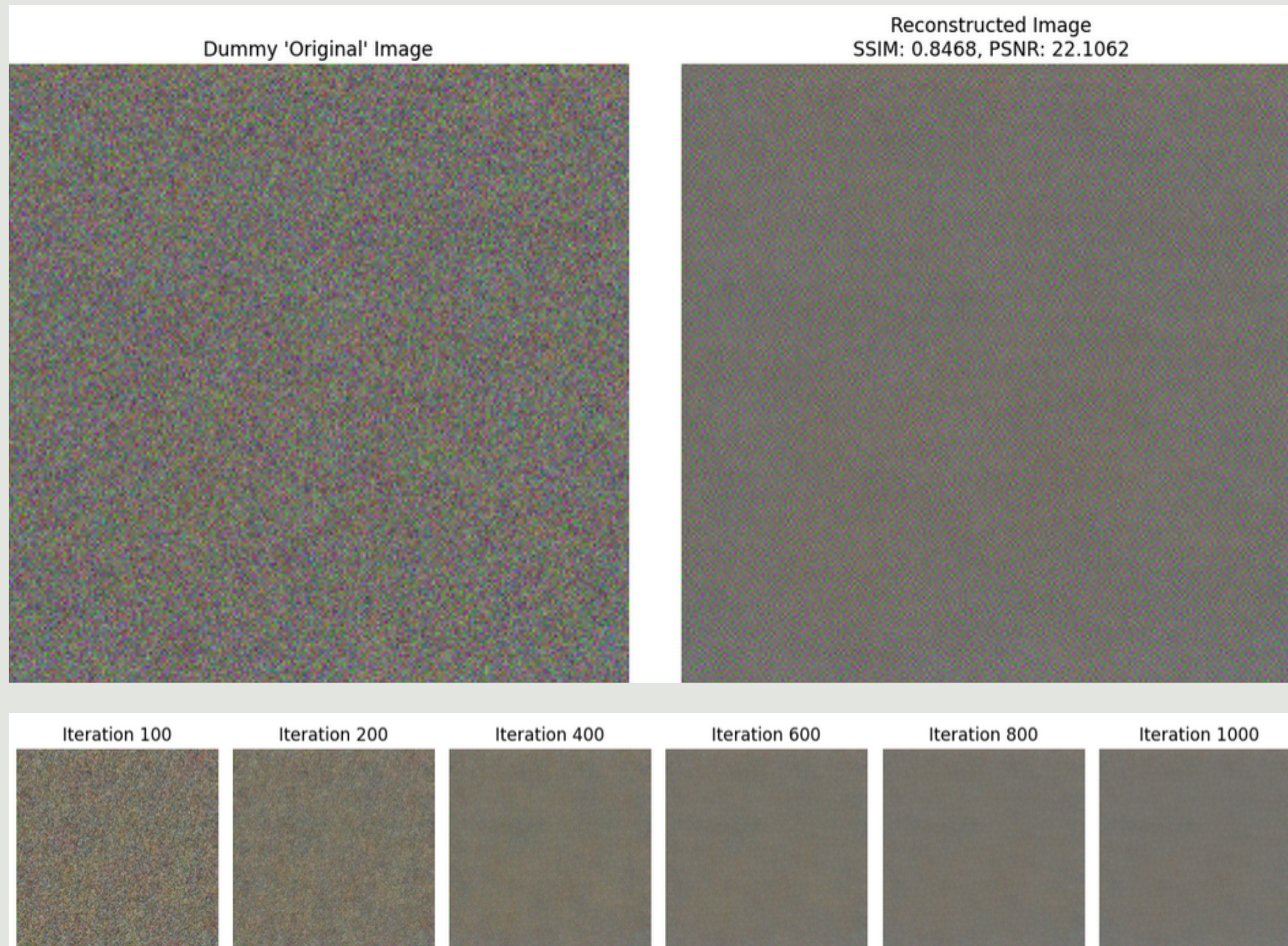
- **OBSERVED OUTPUT:**

- EARLY RECONSTRUCTIONS ARE BLURRY AND PIXELATED
- CAPTURES LOW-LEVEL VISUAL INFORMATION (EDGES, BLOBS)
- LACKS OBJECT-LEVEL SEMANTICS OR TEXTURE DETAIL
- RESULT OF RELYING ON CONV1_1 – CONV2_1 LAYER FEATURES

- **INTERPRETATION:**

- PIXELATED IMAGES ARE VALID OUTPUTS OF LOW-LEVEL RECONSTRUCTION
- THEY REFLECT HOW THE VISUAL CORTEX ENCODES BASIC INFORMATION
- QUALITY IMPROVES AS HIGHER LAYERS (E.G., CONV5_1, FC7) ARE INCLUDED IN OPTIMIZATION

LOW LEVEL RECONSTRUCTION



KEY PITFALLS IN FMRI IMAGE RECONSTRUCTION (OBSERVED ISSUES) SO FAR..

- **OBSERVATIONS FROM REPLICATED MODELS:**
 - RECONSTRUCTIONS ARE BLURRY, NOISY, AND LACKING IN SEMANTIC STRUCTURE.
 - LOW SSIM AND PCC VALUES INDICATE POOR PERCEPTUAL AND STATISTICAL SIMILARITY.
- **CORE PROBLEMS IDENTIFIED:**
 - LIMITED SUBJECT POOL: MOST EXPERIMENTS WERE RUN ON ONLY 1–3 SUBJECTS, LEADING TO POOR GENERALIZATION.
 - SHALLOW FEATURE DECODING: MANY RECONSTRUCTIONS RELY ON LOW-LEVEL VGG LAYERS, WHICH ONLY ENCODE EDGES/TEXTURES—NOT OBJECT IDENTITY.
 - COMPUTATIONAL CONSTRAINTS:
 - FULL FEATURE-SPACE OR END-TO-END LEARNING IS COMPUTATIONALLY EXPENSIVE.
 - REPLICATIONS USED LIGHTWEIGHT MODELS INSTEAD OF LARGE GANS OR TRANSFORMERS.
 - FOURIER WARPING ARTIFACTS:
 - FREQUENCY MISMATCH IN FMRI → PIXEL-SPACE CAUSES DISTORTED STRUCTURES.
 - LACKS TEMPORAL RESOLUTION TO PRESERVE SHARP TRANSITIONS OR SPATIAL PHASE ACCURACY.
- **CONCLUSION: THESE PITFALLS EXPLAIN THE GAP BETWEEN OUR REPLICATIONS AND RESULTS REPORTED IN ORIGINAL PAPERS LIKE SHEN ET AL. (2019).**

OUR NOVELTY – INCORPORATING POST-PROCESSING DENOISING ALGORITHMS

- **OBJECTIVE: TO ENHANCE PERCEPTUAL QUALITY AND SIGNAL COHERENCE OF RECONSTRUCTED IMAGES USING DOMAIN-ADAPTED NOISE REDUCTION TECHNIQUES.**

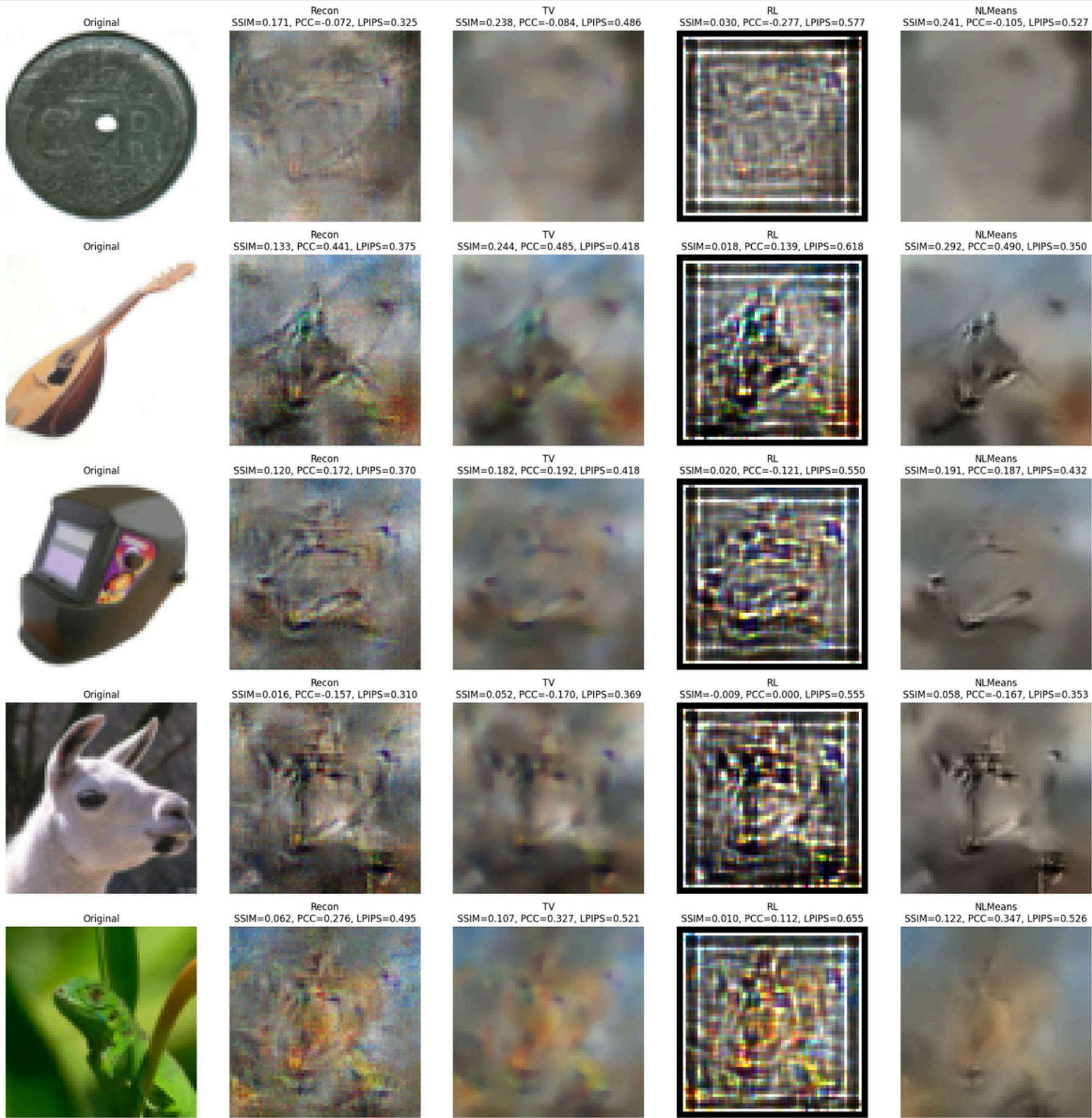
Method	Description	Purpose
TV Denoising	denoise_tv_chambolle	Removes high-frequency noise while preserving edges.
Richardson-Lucy Deconvolution	Iterative PSF-based restoration	Reverses blur caused by convolutional approximations.
Non-Local Means (NL-Means)	Neighborhood averaging	Smoothens similar regions with minimal structure loss.
BM3D	Block-matching 3D filter	Advanced denoising maintaining texture fidelity.

- **EVALUATION METRICS USED:**
 - **SSIM: STRUCTURAL SIMILARITY**
 - **PCC: CORRELATION WITH ORIGINAL**
 - **LPIPS: LEARNED PERCEPTUAL SIMILARITY (DEEP FEATURE DISTANCE)**
- **PIPELINE VISUALIZATION:**
 - **ORIGINAL → RECONSTRUCTION → {TV | RL | NLMEANS | BM3D} → QUALITY EVALUATION**

WHY NOISE REDUCTION MATTERS IN FMRI IMAGE RECONSTRUCTION

- **FMRI-SPECIFIC CHALLENGES:**
 - **BOLD SIGNAL VARIABILITY:** NOISY BY NATURE, WITH LOW SNR AND SLUGGISH HEMODYNAMIC RESPONSE.
 - **VOXEL-LEVEL GRANULARITY:** INDIVIDUAL VOXEL SIGNALS ENCODE AGGREGATE ACTIVITY, NOT CRISP FEATURES.
 - **SPARSE REPRESENTATIONS:** ONLY CERTAIN BRAIN REGIONS ACTIVATE, LEADING TO PARTIAL FEATURE DECODING.
- **WHY POST-PROCESSING IS CRUCIAL:**
 - **ENHANCES SIGNAL CLARITY** IN RECONSTRUCTIONS FROM INCOMPLETE/NOISY FEATURES.
 - **MIMICS PERCEPTUAL SMOOTHING** PERFORMED BY THE HUMAN BRAIN.
 - **IMPROVES DOWNSTREAM METRICS** (SSIM, PCC, LPIPS) AND VISUAL INTERPRETABILITY.
 - **MAKES SEMANTIC DECODING** (E.G., OBJECT DETECTION, CATEGORIZATION) FEASIBLE ON OUTPUTS.
- **CONCLUSION:** DENOISING ISN'T JUST A COSMETIC FIX IT'S ESSENTIAL TO BRIDGE THE MODALITY MISMATCH BETWEEN NOISY BRAIN SIGNALS AND HIGH-RESOLUTION IMAGE SPACE.

Comparative Analysis of Denoising Methods



Key Observations

- Richardson-Lucy (RL):** Highest LPIPS scores (0.577-0.655) indicating strong perceptual quality, but lowest SSIM values (0.009-0.030) showing structural divergence
- Total Variation (TV):** Best balance of metrics with highest SSIM (0.052-0.244) and competitive PCC values (0.084-0.485)
- Non-Local Means:** Superior PCC scores in most cases (up to 0.490), suggesting better statistical correlation with originals
- Raw Reconstruction:** Baseline comparison showing moderate SSIM but inconsistent perceptual quality

Method Characteristics

Reconstruction	TV Filtering	Richardson-Lucy	NL-Means
Baseline output with moderate noise and blur artifacts	Preserves edges while removing high-frequency noise	Enhances edge details but introduces color artifacts	Smooths regions while maintaining structure
SSIM: 0.016-0.171	SSIM: 0.052-0.244	LPIPS: 0.550-0.655	PCC: 0.105-0.490

Metric Analysis

SSIM	PCC	LPIPS
Structural Similarity Index	Pearson Correlation Coefficient	Learned Perceptual Image Similarity
Best Method: TV Filtering	Best Method: Non-Local Means	Best Method: Richardson-Lucy
Indicates preservation of structural information	Measures linear correlation between images	Closest to human perceptual judgment

THE FUNDAMENTAL CHALLENGE

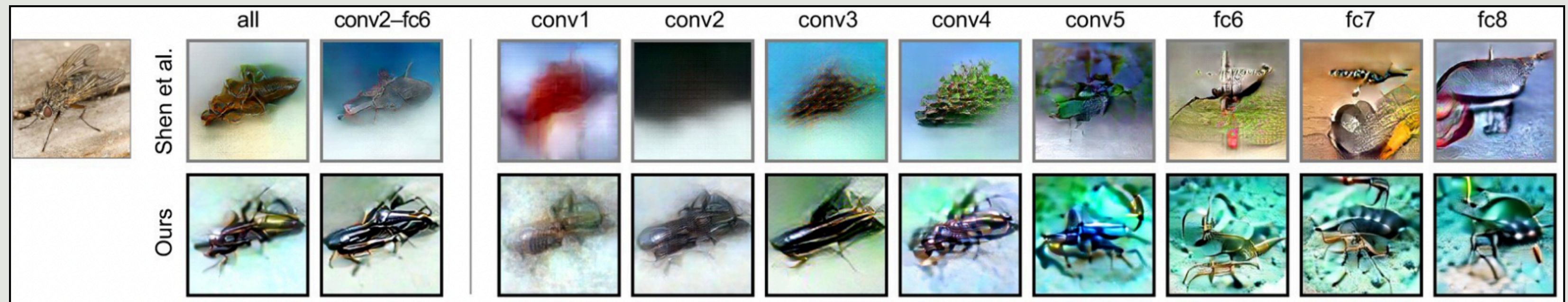
FOURIER WRAPPING IS A PERVASIVE ISSUE IN ALL FREQUENCY-DOMAIN RECONSTRUCTION METHODS USED IN OUR PROJECT:

- **CIRCULAR BOUNDARY CONDITIONS: THE FOURIER TRANSFORM INHERENTLY ASSUMES YOUR IMAGE IS PERIODIC (REPEATS INFINITELY IN ALL DIRECTIONS). WHEN DECONVOLUTION IS APPLIED IN THE FOURIER DOMAIN:**
 - THE LEFT EDGE OF THE IMAGE IS TREATED AS IF IT'S CONNECTED TO THE RIGHT EDGE
 - THE TOP EDGE IS CONNECTED TO THE BOTTOM EDGE
 - THIS CREATES ARTIFICIAL DISCONTINUITIES AT THE BOUNDARIES
- **ORIGIN MISALIGNMENT: MISALIGNMENT OF THE FOURIER ORIGIN CAUSES PHASE SHIFTS THAT MANIFEST AS SPATIAL DISPLACEMENT IN THE RECONSTRUCTED IMAGE. WHEN THE RICHARDSON-LUCY ALGORITHM TRIES TO CORRECT FOR BLUR:**
 - IT AMPLIFIES HIGH FREQUENCIES
 - ANY MISALIGNMENT IN THE FOURIER ORIGIN GETS EXAGGERATED
 - THIS CAUSES PARTS OF THE IMAGE TO "LEAK" OR WRAP INTO OTHER REGIONS
- **SPECTRAL LEAKAGE: THE DISCRETE NATURE OF THE FFT COMBINED WITH THE FINITE IMAGE BOUNDARIES CAUSES ENERGY TO "LEAK" BETWEEN FREQUENCY COMPONENTS, CREATING THE GRID-LIKE INTERFERENCE PATTERNS VISIBLE IN YOUR RESULTS.**

SEMANTIC-BASED RECONSTRUCTION: OUR IMPLEMENTATION (ROUGHLY BASED ON KOIDE-MAJIMA ET AL. (2024))

Why We Chose a Semantic Framework

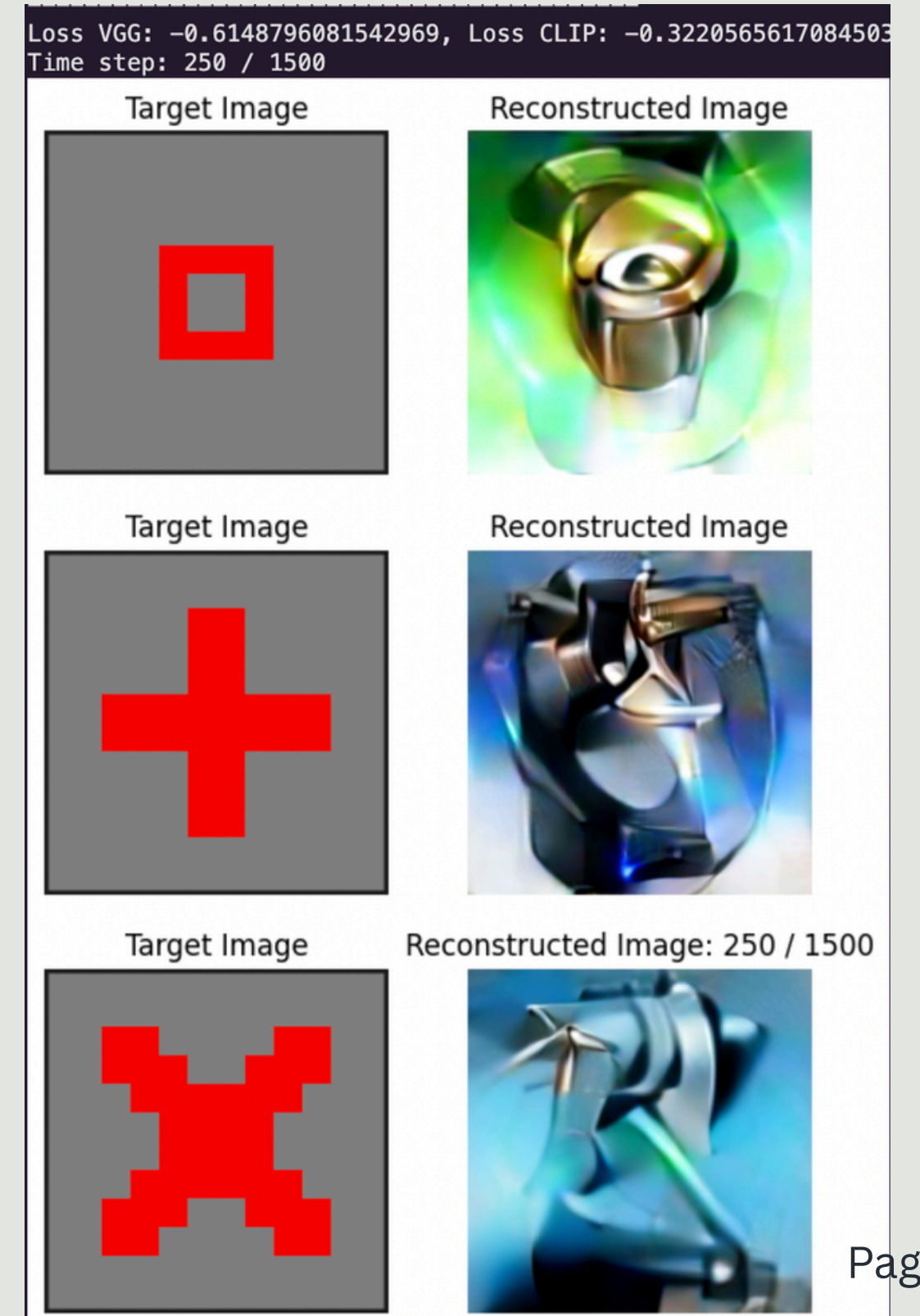
- Initial attempts using Shen et al.'s method showed strong reliance on low-level visual features, which deteriorated under mental imagery and noisy fMRI conditions.
- Mental imagery and abstract concept reconstruction demand models that can decode high-level semantic features from brain activity, not just pixel patterns.
- Semantic priors (e.g., category-level or CLIP embeddings) align better with how the brain encodes complex and imagined content.
- This motivated us to shift from pixel-space reconstruction to a semantically conditioned generation approach.



IMPLEMENTATION PIPELINE

Implementation Pipeline Overview

- **Step 1 – Decoding Stage:**
 - Linear regression models trained to decode:
 - Multi-layer VGG19 features (conv1_2 to fc8).
 - CLIP-ViT-B/32 semantic embeddings from fMRI data.
 - Voxel selection via PCA + correlation ranking to reduce dimensionality and prevent overfitting.
- **Step 2 – Probabilistic Generation:**
 - Image reconstruction formulated as Bayesian posterior sampling over a learned image prior.
 - Used VQGAN pre-trained on ImageNet as a natural image prior generator.
 - Sampling done using Stochastic Gradient Langevin Dynamics (SGLD) to iteratively update the latent vector z .
- **Step 3 – Semantic Guidance:**
 - Incorporated decoded CLIP features into the likelihood function.
 - This allowed the generator to prioritize semantically correct reconstructions, even when visual decoding was weak.
 - Tuned the semantic contribution via a hyperparameter λ CLIP for better imagery performance.



CONCLUSIONS

PROJECT OUTCOMES:

- **SUCCESSFULLY IDENTIFIED LIMITATIONS IN EXISTING FMRI RECONSTRUCTION ARCHITECTURES**
- **IMPLEMENTED BOTH PIXEL-LEVEL (SHEN ET AL.) AND SEMANTIC-GUIDED (KOIDE-MAJIMA) MODELS**
- **DEMONSTRATED SIGNIFICANT QUALITY IMPROVEMENTS VIA POST-PROCESSING TECHNIQUES**
- **ACHIEVED MEASURABLE GAINS IN SSIM, PCC, AND LPIPS METRICS**

BROADER IMPLICATIONS:

- **POST-PROCESSING IS ESSENTIAL FOR ENHANCING NOISY BRAIN-DECODED IMAGES**
- **TOTAL VARIATION FILTERING PROVIDES THE BEST BALANCE OF STRUCTURE PRESERVATION**
- **SEMANTIC FRAMEWORKS MORE EFFECTIVELY RECONSTRUCT ABSTRACT/IMAGINED CONTENT**

REFERENCES

THE BRAINOMICS/LOCALIZER DATABASE

Federated Learning for Medical Image Analysis: A Survey.

Fast Reproducible Identification and large-scale databasin

Natural Image Reconstruction From fMRI Using Deep Learning: A Survey.

Bayesian Reconstruction of Natural Images from Human Brain Activity.

Reconstructing faces from fMRI patterns using deep generative neural networks

FedABIDE: PyTorch implementation of Multi-site fMRI Analysis Using Privacy-preserving Federated Learning and Domain Adaptation: ABIDE Results

Deep Image Reconstruction: Kamitani Lab

VQGAN → Esser, Patrick, Robin Rombach, and Bjorn Ommer. "Taming transformers for high-resolution image synthesis." Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021.

RICHARDSON-LUCY DECONVOLUTION AS A GENERAL TOOL FOR COMBINING IMAGES WITH COMPLEMENTARY STRENGTHS

IMAGE RESTORATION USING TOTAL VARIATION REGULARIZED DEEP IMAGE PRIOR



Thank You

For your attention